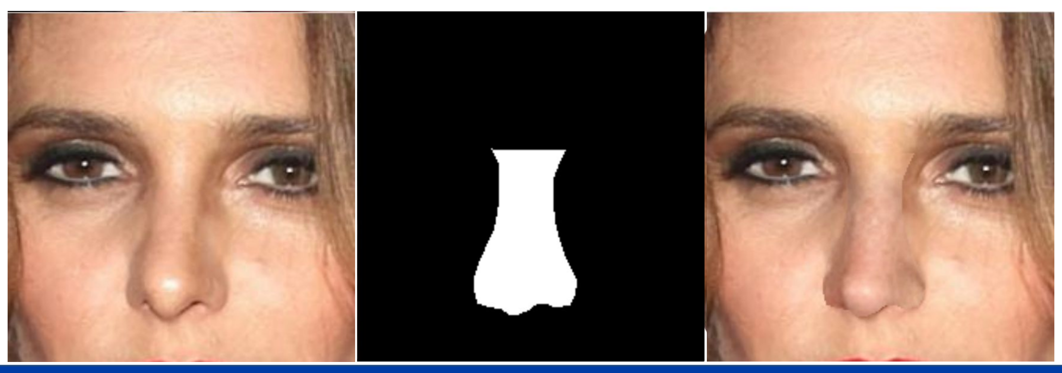
Research Horizons Day
& Research Week
2026

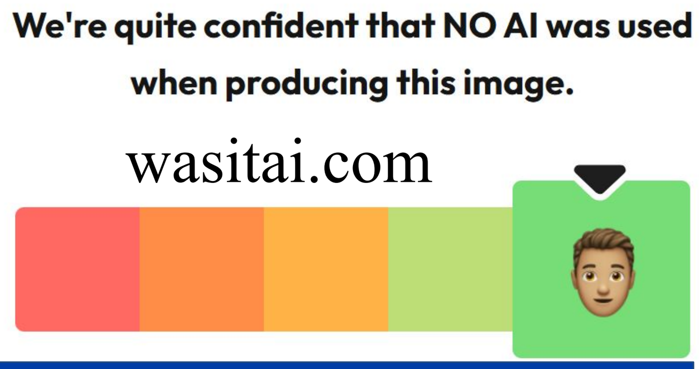
Localizing AI Image Manipulations by Learning Generative Noise Signatures using Diffusion Based Noise Priors

Elijah Shannon, Computer Science Department
Dr. Chaity Banerjee Mukherjee, Computer Science Department

Overview

AI image editing tools can **alter small regions of otherwise authentic photos**, making it difficult for traditional "real vs. fake" detectors to indicate where an image was changed. This poster presents a **two-stage segmentation pipeline** that extracts Diffusion Noise Features (DNFs) via diffusion inversion and combines them with RGB input in a Swin U-Net to localize AI-altered regions.



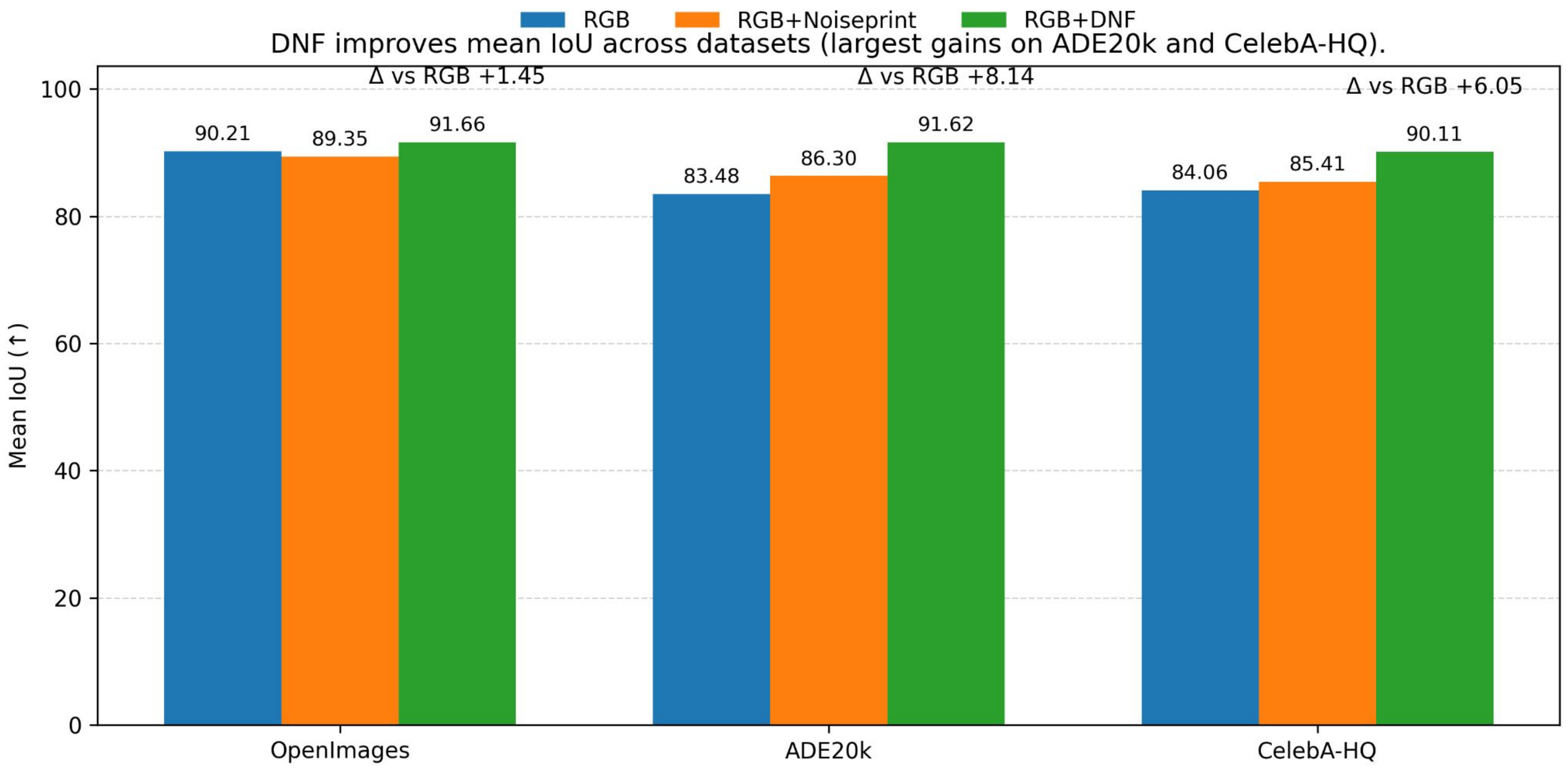


Results

Dataset	Method	Kandinsky 2.2	OpenJourney	SD v4	SD v5	SD XL	Mean
OpenImages	RGB	**92.15**	88.42	88.58	92.8	89.11	90.21
OpenImages	RGB+Noiseprint	87.43	90.24	90.9	90.8	87.36	89.35
OpenImages	RGB+DNF	91.62	**91.01**	**92.26**	**92.97**	**90.46**	**91.66**
ADE20k	RGB	77.88	86.69	81.99	88.77	82.07	83.48
ADE20k	RGB+Noiseprint	86.63	87.73	91.01	81.56	84.56	86.3
ADE20k	RGB+DNF	**93.04**	**91.47**	**90.34**	**91.6**	**91.65**	**91.62**
CelebA-HQ	RGB	85.02	78.44	79.3	90.75	86.81	84.06
CelebA-HQ	RGB+Noiseprint	86.33	83.55	86.73	84.57	85.87	85.41
CelebA-HQ	RGB+DNF	**91.84**	**89.89**	**86.01**	**91.55**	**91.24**	**90.11**

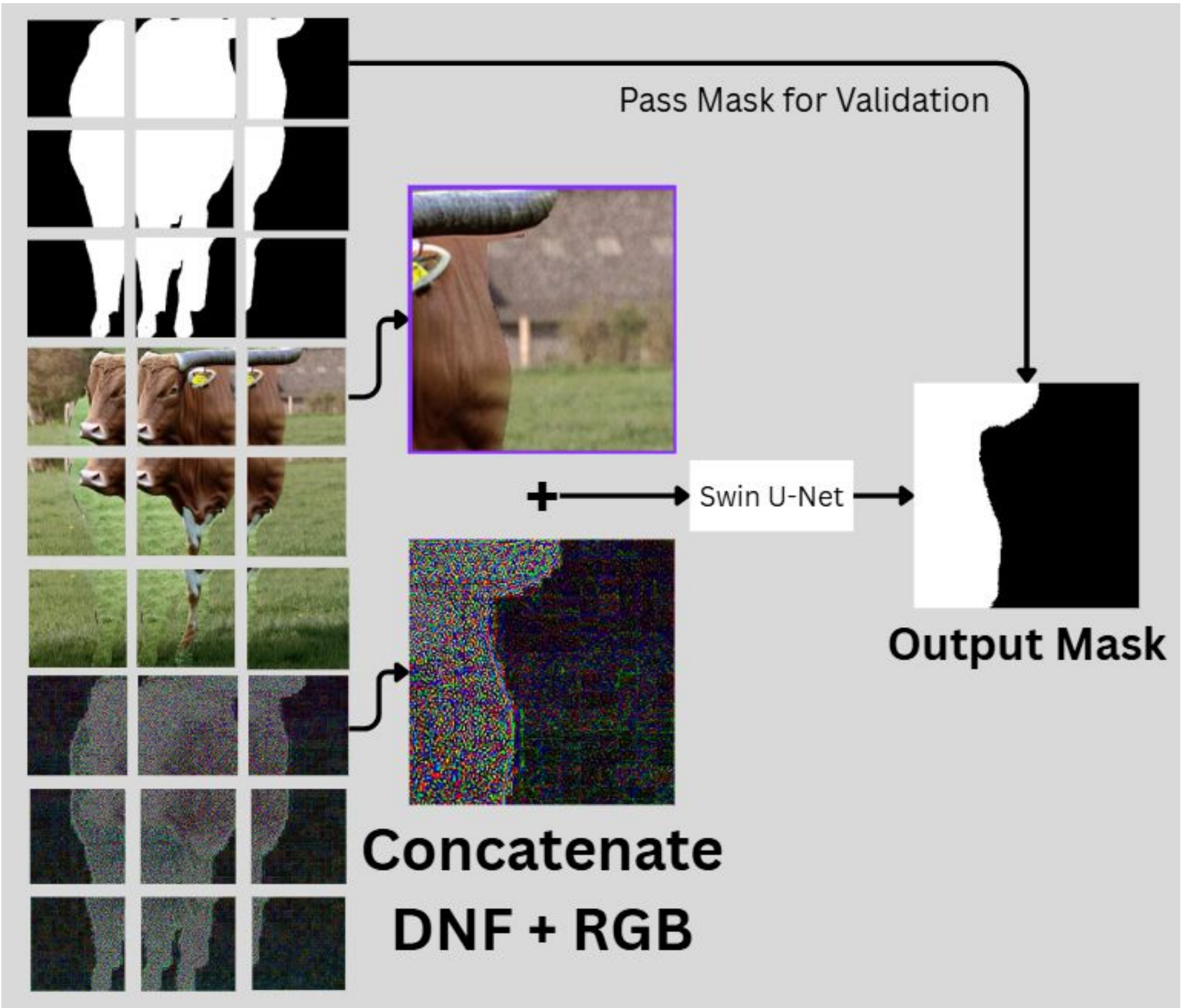
Key takeaway: *RGB+DNF achieves the best mean IoU across all datasets, with the largest gains on ADE20k and CelebA-HQ.*

- DNF improves mean IoU over RGB by **+1.45 (OpenImages), +8.14 (ADE20k), +6.05 (CelebA-HQ)**.
- Noiseprint offers smaller and less consistent improvements compared to DNF.



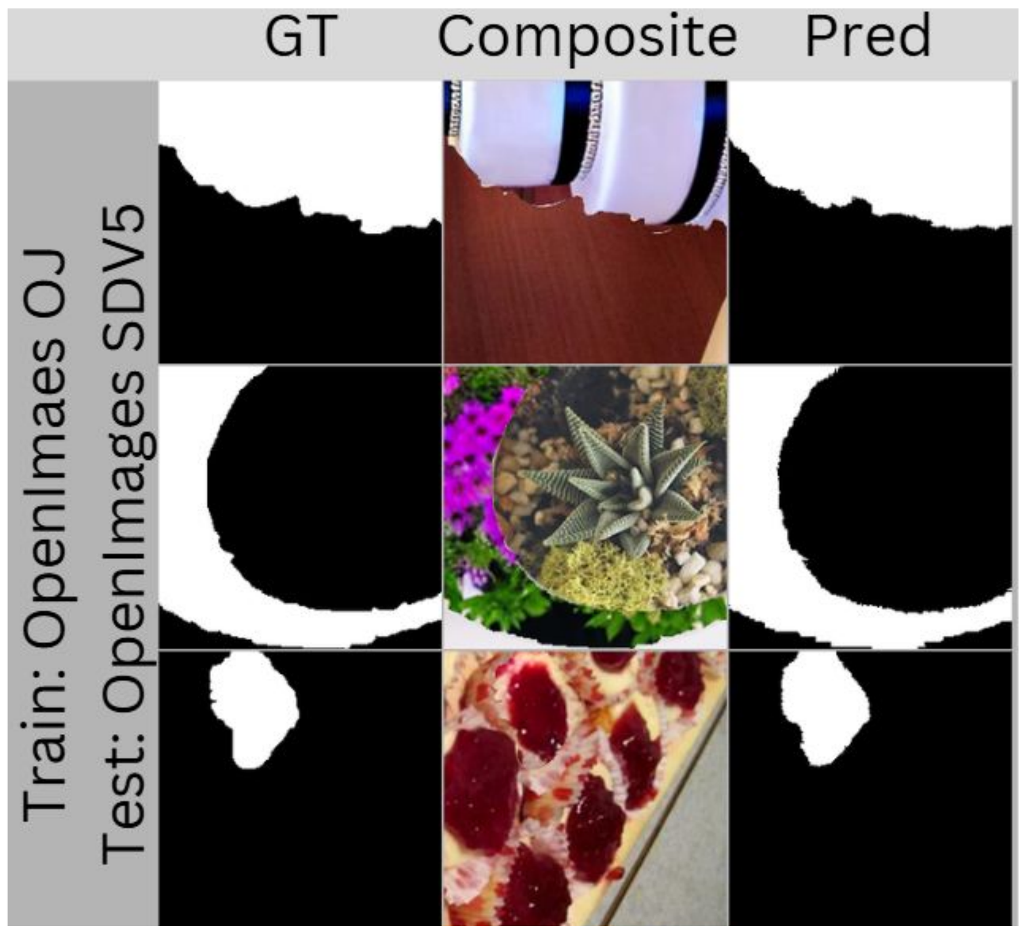
Methodology

1. **Build aligned "composite" training pairs**
   a. Composite = G·M + O·(1−M), producing aligned (original, mask, generated region, composite) quadruplets.
2. **Patch into model-compatible inputs**
3. **Extract Diffusion Noise Features (DNF)**
   a. We use a first-noise residual
4. **Concatenate RGB + DNF to 4-Channel Input**
5. **Train with robust loss + light augmentations**
   a. 0.3·Cross-Entropy + 0.7·Dice



Impact

DNF provides a ***noise-based prior*** that makes manipulation boundaries ***easier to localize***, improving mean IoU and generalizing across diffusion generators producing ***more trustworthy pixel-level masks***.



Partially AI-generated content plays a vital role in:

- Journalism
- Social Media
- Trustworthy Media
- Justice System
- Moderation

This system assists in restoring media confidence.

References

This poster is based on:
Elijah Shannon and Chaity Banerjee, "Localizing AI Image Manipulations by Learning Generative Noise Signatures using Diffusion Based Noise Priors," Applied Artificial Intelligence for Public Safety and Security Workshop (AI-PublicSecurity-2025) at the IEEE International Conference on Big Data (IEEE BigData 2025), Macau, China, December 2025.

COLLEGE OF SCIENCE
THE UNIVERSITY OF ALABAMA IN HUNTSVILLE
THE UNIVERSITY OF ALABAMA IN HUNTSVILLE
HONORS COLLEGE